

## **Bridging Babel: The Career of NLP Specialists and the Quest for Multilingual Knowledge**

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(Pieter Bruegel the Elder, 1563)

### **Introduction:**

According to the [Tower of Babel story in the Book of Genesis](#), a long, long time ago, there was perfect peace. There were no wars, no shortage of food, and accessible knowledge. During this time, there was only one language spoken in the whole world. Trade fostered, and as far as a person could travel, they would still be among their kin. This world was brought about

not through tremendous sacrifice or tedious effort, but through simplicity: the same as it always was.

As the story goes, one day, God decided to change all of this, creating many different languages. These languages brought about misunderstanding, confusion, and separation. Humans no longer looked at one another as friends, but as foreigners. The human world became much smaller, as individuals only loved what they could understand.

Today, with applications such as Google Translate, we understand each other more than ever. People who have nothing in common, from opposite sides of the world, can easily communicate with one another. While this communication is not perfect, through patience and consideration, a common ground can be found. Unfortunately, unlike The Tower of Babel, there is still violence and distrust in the world; however, through consistent improvements in Natural Language Processing (NLP), the world may once again be able to understand one another.

### **The Architects of NLP:**

Those leading this new frontier are not kings or prophets of an ancient world, but NLP specialists with titles just as powerful: [Data Scientist](#), [NLP Engineer](#), and [Computational Linguist](#).

These specialists work in teams where Data Scientists or Data Engineers first collect, clean, preprocess, and annotate large multilingual datasets, inputting the data into a specific model (complex algorithm), and outputting a desired result. Eventually, this process is transformed into a complete pipeline where new data is added, automatically pre-processed, passed into a model, and outputting usable outputs.

### **Dataset Creation:**

Data collection and cleaning are both tedious and essential: only high-quality data can yield quality results (trash in, trash out). Manually labeling data takes considerable time and money. Even then, data quality issues are nuanced and difficult to detect. This is accentuated through the reliance on large datasets, where noise, like special characters or misspellings, can appear, decreasing model performance.

There is a delicate balance between data quality and quantity. Transformer-based models (further explanation provided in the next section) thrive on vast amounts of data, but in specialized tasks like legal translation, precision matters more than scale. This means that when solving an NLP task (or any other task using artificial intelligence), the data collection step must be tailored to the specific use case.

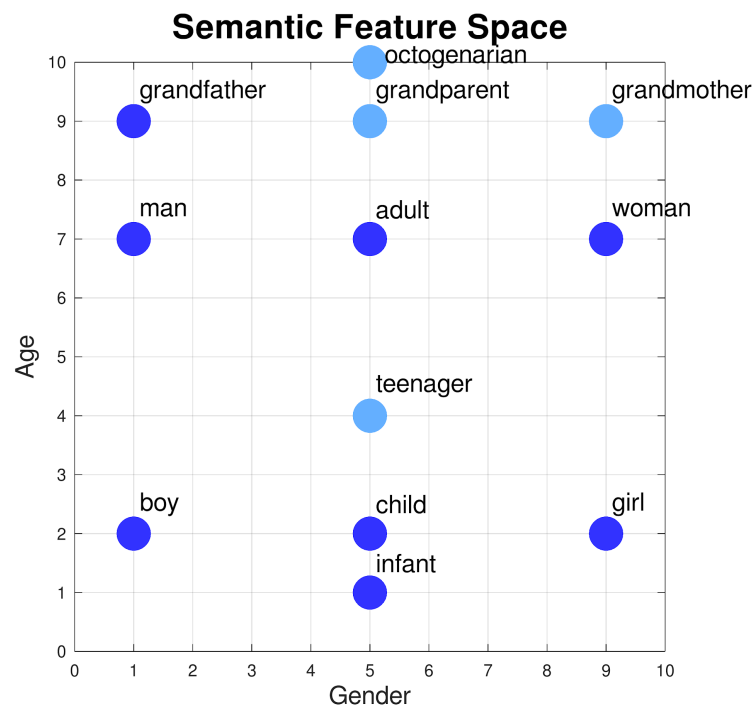
### **Feature Engineering:**

Once the data is ready, several different model architectures are adjusted to learn patterns, distinguish useful signals from noise, and fit the data. Meaningful outputs are guided through feature engineering. In feeding a model data in the form of raw text, a model may recognize incongruent patterns. Even if a model is performing well, the patterns a model learns must be intuitive.

For example, if the goal is to predict the next word in a sequence, a model would learn to make predictions based on probability and sentence length. On the other hand, it is much more useful for a model to make predictions using context or vocabulary cues. For this reason, features such as embeddings, word counts, and part-of-speech (POS) tags are created to direct model learning.

## Embeddings:

Embeddings are numerical representations that capture the semantic meaning of words. These values are mapped in a high-dimensional space, with similar words mapped closer together through model training. The model groups words that appear together or are used interchangeably. For example, the words “apple” and “banana” should be closely mapped; on the other hand, words such as “Tetris” and “bedtime” should be more distant. Embeddings can be very useful to better understand word similarity, and by generating visualizations we are given insight into how models learn conceptual relationships through embeddings.



**The visualization above shows how family-related words cluster in a semantic feature space based on gender (x-axis) and age (y-axis). Words like grandfather and grandmother occupy similar regions, while boy and girl cluster at lower ages (Jason Xu, 2022).**

**TF-IDF:**

Next, word-count features such as TF-IDF (Term Frequency–Inverse Document Frequency) measure how often a word appears in a specific document (TF) while reducing the weight of words that frequently appear across many documents (IDF). This allows TF-IDF to highlight words that carry unique or meaningful information rather than common filler words: like, the, or, and and.

This technique is especially useful in semantic classification tasks, where the presence of a rare but meaningful word can signal a specific tone or intent. For example, a marketing slogan using a distinctive word like “indubitably” (as famously used in a [Crispy Critters Cereal campaign](#)) stands out because it carries strong emotional emphasis and is rarely seen in everyday language.

TF-IDF gives such uncommon terms more weight, helping the model recognize that they convey special significance. By valuing unique vocabulary, TF-IDF encourages models to better understand nuanced language and build a more robust language representation.

**POS tags:**

Lastly, POS tags (adjective, noun, punctuation, etc.) provide syntactic context, helping models understand not just what a word means, but how it functions in a sentence. This context is vital for nuance, sentiment, and meaning: the very foundation of true language understanding.

By recognizing the role each word plays, models are less likely to misinterpret meaning, especially in sentences where the same word can appear in different forms. For instance, POS tagging helps distinguish between “run” as a verb (“I run every morning”) and “run” as a noun (“He went for a run”). This reduces ambiguity, improves sentence interpretation, and supports more accurate downstream NLP tasks.

## Understanding language:

The steps taken for a model to understand a language such as English are very complex: a machine must understand both syntax and semantics, all of which can change given context. In translation to another language, such as Mandarin, this same understanding is required through varying order of words, length, or translations.

Throughout the NLP workflow, linguists are vital to ensure proper understanding of language for the generation of meaningful results. Specifically, linguists are involved mainly in dataset design, annotation, and evaluation. They examine example outputs, provide expert-corrected translations, and build gold-standard test sets. They don't manually teach grammar rules, but ensure training data and evaluation metrics reflect real language use, idioms, and cultural nuance.

Their feedback shapes model behavior during fine-tuning and error analysis, ensuring the system learns meaningfully and avoids literal mistranslations. To illustrate the difficulties of translation, the example below shows a sentence in Mandarin, given its direct word definition translation, and then its actual translation.

梅 干 菜          用 清 水 泡 软 , 捞      出 后 , 沥 干  
plum dry vegetable use clear water soak soft , remove out after , drip dry  
切    碎  
chop fragment

*Soak the preserved vegetable in water until soft, remove, drain, and chop*

**While word definitions can be understood and translated, to make the translation grammatically correct and cohesive for broader understanding, additional steps are required. A linguist ensures effective model translations through continual evaluation (5, Speech and Language Processing).**

## **Modeling using Transformers:**

Next, NLP Engineers and Machine Learning specialists take the preprocessed, annotated data to train and fine-tune models. This work is done in iterations to improve accuracy, fairness, or usability for their specific use case. In the last five years, the unique architecture of Transformers, first proposed in 2017, has become the foundation of NLP. [Decoder-based transformers](#) such as ChatGPT have revolutionized how millions around the world learn, communicate, and perform their jobs.

## **Attention:**

Transformers learn through attention: the ability of the model to understand words and their relationship with one another. Attention is used for the specific use case of the model through three parts:

1. Query: What am I looking for?
2. Key: What do I contain?
3. Value: What should I return?

The core of a Transformer lies in its multi-head attention blocks. Each head is a specialist of a different linguistic phenomenon (syntax, semantics, coherence, etc.) deciding what information to pass on.

## **XAI:**

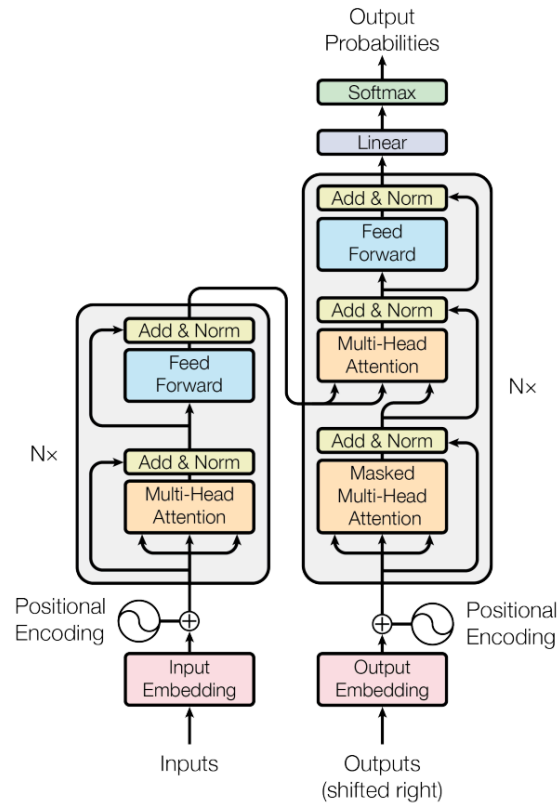
Another large part of Machine Learning is Explainable AI (XAI). Models like Transformers are considered black box models. This means that it is impossible to truly understand how the model makes decisions. XAI is a method used to better understand how models make decisions, creating stronger ethical alignment.

**Architecture:**

After the attention step, a feed-forward network is used to transform this information into usable outputs. One of the Transformer's greatest strengths is parallelization. Instead of processing text one word at a time, it processes many relationships simultaneously. This design makes training much faster and more efficient than sequence models like [RNNs](#) or [LSTMs](#).

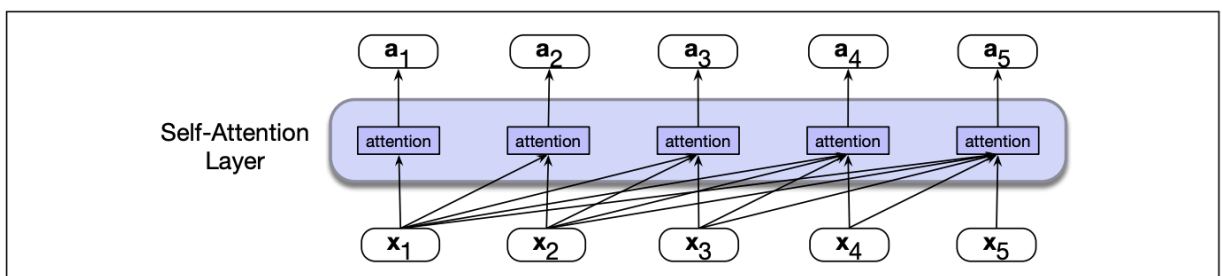
Below is the original architecture suggested in the paper [“Attention Is All You Need” by Vaswani et al. \(2017\)](#). On the left side is The Encoder. Models take in the input sentence, break it down into numbers the model can understand (through embeddings), to learn relationships between words using attention. On the right side is The Decoder, first taking in what the encoder understood, using that knowledge to predict the next word in another language. Each new predicted word helps it guess the next one better, and the learning of one language aids the learning of all languages (cross-lingual learning).





## Context Windows:

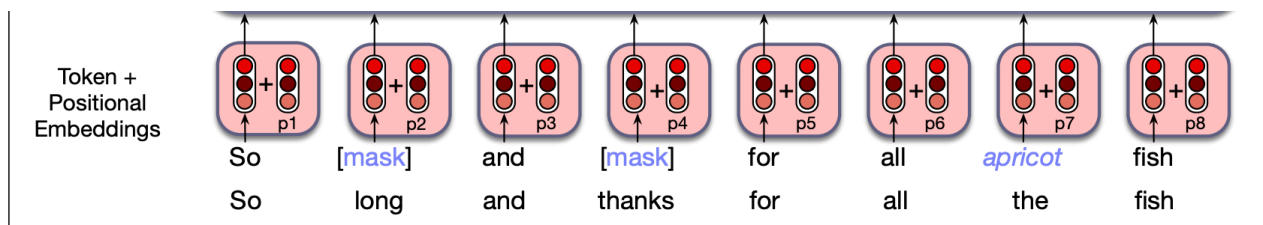
A majority of transformer models sequentially feed the data using a context window. This [parameter](#) (changeable feature of the model) is used to select how many of the words before the current word are used to predict the next word.



**Figure 8.3** Information flow in causal self-attention. When processing each input  $x_i$ , the model attends to all the inputs up to, and including  $x_i$ .

## Masking:

On the other hand, masking is also used. Masking takes a large document and masks words at random, given a subset of the data (150 words). Within this subset, there are several words hidden from the model. The model uses a different window: the context of words in front or behind the word for predictions. Here, the model can learn context more efficiently by performing only a few predictions, but learning context for larger subsets.



**Figure 10.3** Masked language model training. In this example, three of the input tokens are selected, two of which are masked and the third is replaced with an unrelated word. The probabilities assigned by the model to these three items are used as the training loss. The other 5 tokens don't play a role in training loss.

## Use Cases and Global Impact of NLP:

Now, through understanding the inner workings of the NLP pipeline, from data collection to model training, we can look outward. What do these efforts actually achieve?

The power of NLP lies not only in the algorithms themselves, but in their ability to transform how people access information and communicate across linguistic and cultural boundaries. Specifically, three major efforts in NLP are expansion of information globally, creation of specialized applications, and bridging the gap for low-resource languages.

## Access to Information:

To begin, tech giants such as Google, Microsoft, Meta, OpenAI, and Amazon offer free tools (Google Translate, Microsoft Translator) to expand reach and drive users to their ecosystem (ads, cloud services, premium products). Through these translation services, travelers in any

country can ask a local where the nearest pub is, farmers can receive weather information in their native language, or a child can read their favorite Comic Book.

Additionally, the same mechanisms that translate text also power sentiment analysis, content moderation, speech recognition, topic modeling, trend detection, information extraction, and personalized AI assistants and chatbots. In all of these efforts, the understanding of language and the accuracy of translation are improved. Work in each sector of NLP interacts reinforces the others, reshaping how humans interact with information, languages, and even each other.

### **Specialized Applications:**

Specifically, startups are focused on niche NLP applications: medical translation, finance, legal documents, and low-resource languages. These companies make money by licensing AI models, selling analytics, or providing enterprise solutions. In medicine, NLP systems translate and summarize complex health documents, ensuring clarity and safety for patients. In finance, models detect fraud and analyze global market sentiment. In law, they streamline document review and contract analysis. In each domain, NLP specialists tailor models to specific linguistic contexts, collecting and refining domain-specific datasets to help systems interpret specialized vocabulary accurately.

### **Underrepresented Languages:**

This leads to perhaps the most important and challenging issue within NLP: inclusion of underrepresented languages. Thousands of languages lack the data needed to train effective models, leaving many communities excluded from digital access. NLP's goal is to reach the furthest communities, providing the benefits of translation to everyone. So, how do NLP

specialists create accurate translations for languages without data? The answer: more NLP.

Individuals in the most remote parts of the world, speaking uncommon languages, are entitled to access to information. Through these initiatives in NLP: creating new datasets, developing transfer learning techniques, and supporting grassroots data collection efforts, individuals in remote regions are not left behind by technological progress.

### **Conclusion:**

Overall, NLP is providing digital access to communities once left behind by the dominance of English online, breaking patterns of linguistic assimilation, and building a stronger, more connected world community.

Through increased efforts in NLP, from corporate innovation to local data set creation, the world will begin to understand one another again. Through language, technology, and collaboration, NLP holds the potential to create not only smarter systems but a more peaceful and inclusive future.

**If you would like to learn more about current efforts to make NLP more inclusive for underrepresented languages, explore the following projects:**

[Masakhane Project](#) – A grassroots research community building open-source NLP tools for African languages.

[Mozilla Common Voice](#) – A global initiative to create open voice datasets and support low-resource languages.

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